MILITARY SPECIFICATION

MICROSCOPE, SURGICAL, Co-observation, Binocular 120/230 Volt, 50-60 Hertz, AC

This limited coordinated military specification has been prepared by the Defense Personnel Support Center, based on currently available technical information. However, pending its promulgation as a coordinated military specification, it may be used in procurement.

1. SCOPE

1.1 Scope. This specification covers requirements for a surgical microscope with a reusable container.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

Military

DPSC-DEPMEDS-AT(DM)	-	Deployable Medical Systems, General Requirements for.
MIL-D-42048	-	Deployable Medical Systems, General Requirements for.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-RST, 2800 South 20th Street, Philadelphia, PA 19101, by using the self- addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

(DD Form 1426 to be included in the final document.)

AMSC N/A FSC 6515

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

3. REQUIREMENTS

3.1 General. The complete requirements for acquiring the surgical microscope described herein shall consist of this specification and the latest issue of Military Specification MIL-D-42048, except where otherwise indicated.

3.2 Design Characteristics. Shall be a fiber optic illuminated and motorized operating microscope with interchangeable objectives for use in ENT, Ophthalmic and Neurosurgery.

3.2.1 Components. As a minimum, shall include the following:

A. A microscope body with motorized zoom magnification changer, focus adjustment, and body positioning; integrated coaxial fiber optic illumination with a minimum 60-inch lightguide, and interchangeable 175 mm, 250 mm and 300 mm objectives.

B. A fiber optic coaxial illumination system which includes an illuminator housing with built-in power supply, lamp on/off and intensity control; enough light output ports to transmit light to the main microscope illumination system, slit illuminator and homogeneous illuminator, and any necessary column mounting brackets.

C. Inclined binocular tube equipped with removable high-eyepieces with locking diopter positions, and a continuously adjustable eyepiece interpupillary distance of approximately 50 mm to 75 mm.

D. A total system magnification ratio of at least 1:4, with a maximum magnification of at least 16X, and a field of view ranging from at least 13 mm to 50 mm.

E. A mobile floorstand that contains base, column and horizontal arm assemblies, and any necessary counterweights and stabilizer weights to maintain safe balance during use. The height of the column shall be at least six (6) feet (1.83 m).

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F. A motorized control system consisting of a foot control panel, power control unit(s) and all necessary connectors to control, at a minimum, magnification changer zoom, focus adjustment, and microscope x-y axis positioning with automatic recentering.

G. External slit and homogeneous illuminators which are prefocused for the working distance provided by the main microscope objective lens, and allow a means for repositioning and refocusing should misalignment or unfocusing occur.

H. Assistant's inclined binocular stereo microscope to assist primary surgeon in surgical procedures; with either zoom or minimum 3-step magnification changer, an overall magnification range of at least 3.6x to 8.9x, and high eyepoint eyepieces with locking diopter positions.

I. Beam splitter for connecting and providing the assistant microscope coaxial observation with the primary field of view (in order to assist primary surgeon in surgical procedures). Adding the beam splitter shall result in the following:

(1) The primary and assistant's fields of view being at the same intensities.

(2) The light intensity to the point of focus does not change, although the amount of light at both pairs of eyepieces may be reduced.

(3) There is no induced prism.

J. Protective cover for the microscope.

K. Four (4) sterilizable sleeves for the inclined binoculars.

L. Two (2) sets of any necessary cable clamps to keep motor control cables out of a sterile field.

M. One spare plug-in module with bulb for the illumination system.

N. Motorized or manual body tilt mechanism that does not restrict the focusing range.

O. Motorized focusing range of at least 50 mm at a speed of 3 mm/second.

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3.3 Maintenance Support.

3.3.1 Provisioning technical documentation. The technical documentation which forms the appendix of DPSC-DEPMEDS-AT(DM) shall be required, and shall be as indicated on the DD forms 1949-1, 1949-2, 1423 and 1664. However, these requirements may be altered or deleted by a solicitation or contract that is applicable to a particular procurement.

3.4 Workmanship. The surgical microscope and reusable container shall be free from defects which detract from their appearance or impair their serviceability.

Unit. Each (EA). One microscope system, as specified constitutes one unit.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used herein, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 First article inspection. First article inspection shall be performed on the surgical microscope when a first article sample is required. This inspection shall include examination and tests.

4.1.4 Quality conformance inspection. Quality conformance inspection shall include the examination and tests specified herein.

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4.1.5 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the component/material, the lot number and the lot size.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table I. The unit of product for sampling purposes shall be one surgical microscope.

Table I. Sampling for examination.

	Inspection level	AQL (percent defects)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5

4.2.2 For tests. Sampling for tests shall be conducted as follows:

a. First article. Where first article sample is required, all tests shall be performed on that unit.

b. Acceptance testing. When a first article sample has not been required, one unit shall be subjected to all tests specified herein. If a first article sample has successfully passed all tests, performance testing shall not be required at acceptance.

4.3 Examination. The surgical microscope shall be examined to determine compliance with the requirements of this specification in accordance with the classification of defects in table II.

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TABLE II. Categories and Defects

Categories	Defects
Major	
101	Components of surgical microscope not free of damage or distortion.
102	Interchangeable objectives cannot be properly secured to or removed from microscope body.
103	Interpupillary distances on inclined binocular tube not properly adjustable.
104	Optical components not free of cracks, chips, striae, or cluster(s) of scratches.
105	High-eyepieces not capable of being secured to or being removed form inclined binocular tube.
106	Floorstand assembly not able to be free of rocking or wobbling when resting on a horizontal level surface.
107	Line cord and cables not free of cuts, cracks or fraying.
Minor	
201	Identification marking not complete, permanent, and legible.
202	Finish not free of chips, cracks, scratches, and other appearance defects.
203	Any component part not secured in place.

Examination shall not be limited to the classified possible defects listed herein.

4.4 Tests. Tests shall be conducted to determine compliance with this specification.

4.4.1 Performance. The surgical microscope shall be completely assembled in its operating configuration and tested to determine compliance with this specification. Testing shall include checking for proper operation of:

A. The motor control system with magnification changer zoom, focus adjustment, x-y axis positioning.

B. System magnification adjustment.

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C. Illumination for coaxial, slit, and homogeneous illuminator.

D. Adjustable focusing of the slit and homogeneous illuminators for the 175 mm, 250 mm and 300 mm working distances.

The surgical microscope assembly shall be electrically operated to determine proper functioning. The surgical microscope shall not be defective.

5. PACKAGING

5.1 General. The complete requirements for packaging shall consist of this specification and the latest issue of Military Specification MIL-D-42048 except where otherwise indicated.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) National Stock Number (NSN).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2)
(d) Exception(s) to requirements in paragraph 3.1 of this specification.

6.2 Stock listing. This specification covers the following item listed in the Federal Supply Catalog:

NSN	ITEM IDENTIFICATION
6515-01-181-1405	MICROSCOPE, SURGICAL, Co-observation Binocular, 120/230 Volt, 50-60 Hertz, AC

6.3 Qualification. This specification does not cover all types, classes, grades or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet military requirements.

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing activity:

DoD-MB

Agent:

DLA-DM

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CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6515-4241

Location: NCR DISK #4

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NOTE. This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-42053 (DM)	MICROSCOPE, SURGICAL, Co-observation, Binocular 120/130 Volt, 50-60 Hertz, AC

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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